Non-Conductive PEN Spray Can

Cost Analysis per Unit — CSMFAB 018

Category: Aegis Home — Hardware **Unit Basis:** per 12-pack (150mL cans) **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

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1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

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2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
PEN (polyethylene naphthalate) bottle preform (12pc)	0.72 kg	$8.00/kg	$5.76
Spin-weld PEN closure (12pc)	0.12 kg	$8.00/kg	$0.96
Metal-free bag-on-valve assembly (12pc)	0.36 kg	$4.00/kg	$1.44
Bio-derived CO2 propellant (12pc)	0.18 kg	$12.00/kg	$2.16
PEEK peristaltic valve mechanism (12pc)	0.12 kg	$280.00/kg	$33.60
YInMn Blue printed label	0.01 kg	$20.00/kg	$0.20
BOM Subtotal			**$44.12**

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3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	$44.12	$36.18	$28.68
Direct Labor	$18.00	$12.00	$8.00
Overhead (15% of labor)	$2.70	$1.80	$1.20
Quality Control & Testing	$2.16	$0.96	$0.40
Packaging & Logistics	$25.00	$20.00	$15.00

Tooling & Equipment Notes: PEN blow molding NRE: $22,000. Standard aerosol filling equipment modified for PEN.

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4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	**$120**	$192 – $264	35-55%
Small Batch (100-500/yr)	**$82**	$123 – $164	33-50%
Production Volume (5K-50K/yr)	**$58**	$81 – $104	28-45%

Market Reference: $72-110/12-pack = $6-9/can (premium vs. $2/can steel; 3× markup for GIC-safe, non-incendiary)

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5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder $350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — $2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; $12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

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6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** $7 – $15 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

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End of Non-Conductive PEN Spray Can Cost Analysis | CSMFAB 018 | June 2026